

The Dissolved Helix and Its Orthogonal Partner

Terrain, rotation, and the completion of $\mathbb{Z}/4\mathbb{Z}$

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Abstract

The self-action of the golden companion R ($R^2 = R + I$) is a single real-axis mode: its commutator spectrum is $\{0, \pm\sqrt{5}\}$ and its angle charge lies in the sublattice $\{0, 2\} \subset \mathbb{Z}/4\mathbb{Z}$. We show, entirely from the exact structure, that this mode cannot *couple to a copy of itself* into a bound state — the generative grade $[R, R]$ vanishes, self-composition keeps the charge in $\{0, 2\}$ while the Mahler measure grows without ever returning to the cyclotomic floor, and the “on-circle-and-expanding” (Salem) channel is empty. Closure requires the orthogonal charges $\{1, 3\}$, and these are supplied by exactly one seed: the quartic $K = x^4 + 5x^2 - 5$, whose real roots $\pm K$ are the *terrain* (in the field $\mathbb{Q}(5^{1/4}) \supset \mathbb{Q}(\sqrt{5})$) and whose imaginary roots $\pm i\beta$ are a $\frac{\pi}{2}$ *rotation about the origin*. The discriminant flip $D = 1 + 4C$ makes terrain and rotation the two faces of one event; the golden strand is the $D > 0$ face and K straddles the fold, re-adjointing the $D < 0$ face. Every result carries an epistemic tag and, where it exists, the repository file that checks it; all displayed constants are machine-verified in exact arithmetic. The “helix” reading is a **[POSITED]** overlay on a **[FORCED]** substrate — but the terrain/rotation anatomy it rests on is the repository’s own. Finally we abstract the procedure that produced these results: a *harness* that imports an external structure, adjudicates each of its claims against the exact substrate, and partitions them into survivors, casualties, and residue (§8). The helix is its first worked instance; the transferable object is the harness, not the helix.

1 Epistemic conventions

Claims are tagged. **[FORCED]**: a theorem of the exact structure, machine-checked in the pipelines. **[COMPUTED]**: an exact fact derived here (symbolic, no floating point crosses a decision). **[ESTABLISHED]**: standard external mathematics. **[POSITED]**: an interpretive overlay (the “helix” language), grounded but not load-bearing. **[OPEN]**: acknowledged open. *No float crosses a decision boundary*: every constant below is decided over $\mathbb{Q}$, $\mathbb{Q}(\sqrt{5})$, or $\mathbb{Q}(5^{1/4})$. Appendix A lists the checks. A claim the substrate *falsifies* (its negation forced by an exact counterexample) is marked **[REFUTED]**; equivalently **[REFUTED]** $\gamma \equiv$ **[FORCED]** $\neg\gamma$.

2 The keystone: the golden generator

Definition 2.1 (Golden data). Let $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2} = 1 - \varphi = -\varphi^{-1}$, the roots of $x^2 - x - 1$. Let $R = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ be the Fibonacci companion.

Proposition 2.2 (The keystone). $\text{charpoly}(R) = \lambda^2 - \lambda - 1$, so $\text{Tr } R = 1$, $\det R = -1$, $\text{spec}(R) = \{\varphi, \psi\}$, and $R^2 = R + I$. Moreover $\varphi\psi = -1$, $\varphi + \psi = 1$, and

$$\varphi - \psi = \sqrt{5} = \varphi + \varphi^{-1}. \quad (1)$$

[FORCED] (drop-one keystone appendix of the $\lambda = 2c$ paper; `test_p1_05.keystone.py`).

Proof. $\det(\lambda I - R) = \lambda(\lambda - 1) - 1 = \lambda^2 - \lambda - 1$, whose roots are φ, ψ ; Vieta gives $\varphi + \psi = 1$, $\varphi\psi = -1$. Cayley–Hamilton yields $R^2 - R - I = 0$. Finally $\varphi - \psi = \frac{(1+\sqrt{5})-(1-\sqrt{5})}{2} = \sqrt{5}$, and $\varphi + \varphi^{-1} = \varphi + (\varphi - 1) = 2\varphi - 1 = \sqrt{5}$. $\square$

Proposition 2.3 (The self-action). *Let $\text{ad}_R = [R, \cdot]$ act on $M_2(\mathbb{R}) \cong \mathbb{R}^4$. Then*

$$\text{spec}(\text{ad}_R) = \{0, 0, +\sqrt{5}, -\sqrt{5}\} = \{0, \pm\sqrt{5}\}, \quad (2)$$

and $\sqrt{5}$ (the sole irrational in the spectrum) is the coupling $\varphi + \varphi^{-1} = \varphi - \psi$. [FORCED] (*test_p2_07_uniform.py*, Lem. 8.1).

Proof. For a diagonalisable R with eigenvalues μ_1, μ_2 , the operator $[R, \cdot]$ has eigenvalues $\{\mu_i - \mu_j\}_{i,j}$ on M_2 . Here $\{\varphi - \varphi, \varphi - \psi, \psi - \varphi, \psi - \psi\} = \{0, \sqrt{5}, -\sqrt{5}, 0\}$ by (1). $\square$

Definition 2.4 (The two characters). An *object* is a finite multiset $A = \{\lambda_1, \dots, \lambda_d\}$ of nonzero algebraic integers (an eigenvalue multiset). Its two characters are

$$M(A) = \prod_i \max(1, |\lambda_i|) \quad (\text{Character I, magnitude}),$$

$$\chi(A) = \left\{ \left\lfloor \frac{2 \arg \lambda_i}{\pi} \right\rfloor \bmod 4 \right\} \subset \mathbb{Z}/4\mathbb{Z} \quad (\text{Character II, phase}).$$

The *golden object* is $A_\varphi = \text{spec}(R) = \{\varphi, \psi\}$, with $M(A_\varphi) = \varphi$ (since $|\psi| = \varphi^{-1} < 1$) and $\chi(A_\varphi) = \{0, 2\}$ ($\varphi > 0 \Rightarrow 0, \psi < 0 \Rightarrow 2$). The operators are $\oplus$ (multiset union; M multiplies, χ unions), $\otimes$ (products $\{\lambda\mu\}$; χ sums mod 4), and the Adams power ψ^2 (squaring; $M \mapsto M^2, \chi \mapsto 2\chi \bmod 4$). [FORCED] (S1–S6).

3 The self-coupling obstruction

We now show the golden mode cannot bind to a copy of itself. Three independent facts converge.

Proposition 3.1 (Cayley–Hamilton collapse). *Let $\Phi_R(Y) = Y^2 - \text{Tr}(R)Y + \det(R)I$ be R 's characteristic operator. Then*

$$\Phi_R(R) = R^2 - R - I = 0, \quad [R, R] = 0. \quad (3)$$

Self-application under R 's own seed law returns the golden recurrence $R^2 = R + I$ and vanishes; the generative grade (the commutator/wedge $[R, R]$) is identically zero. [FORCED] (*KL_DTA.py*).

Proof. $\Phi_R(R) = R^2 - 1 \cdot R + (-1)I = R^2 - R - I = 0$ by Proposition 2.2. $[R, R] = R^2 - R^2 = 0$ by antisymmetry of the Lie bracket. [ESTABLISHED]/[FORCED]. $\square$

Lemma 3.2 (The magnitude never returns to the floor). $M(A \oplus A) = M(A)^2$ and $M(\psi^2 A) = M(A)^2$. Since $M(A_\varphi) = \varphi > 1$, every self-composite has $M \in \{\varphi^2, \varphi^4, \dots\}$; the cyclotomic locus $M = 1$ is never reached. [FORCED] (S4; S10: $M(\psi^2 A) = M(A) \iff M(A) = 1$).

Proof. $\oplus$ concatenates multisets, so M multiplies; ψ^2 squares each λ , so each $\max(1, |\lambda|)$ squares. Iterating $\varphi > 1$ diverges from 1 monotonically. $\square$

Proposition 3.3 (The charge sublattice is closed). *The golden charge $\{0, 2\}$ is invariant under $\oplus, \otimes, \psi^2$ and never reaches $\{1, 3\}$:*

$$\{0, 2\} \cup \{0, 2\} = \{0, 2\}, \quad \{0, 2\} + \{0, 2\} \bmod 4 = \{0, 2\}, \quad 2 \cdot \{0, 2\} \bmod 4 = \{0\}.$$

[FORCED] (S6).

Proof. Direct: $0 + 0 = 0$, $0 + 2 = 2$, $2 + 2 = 4 \equiv 0$; and $2 \cdot 0 = 0$, $2 \cdot 2 = 4 \equiv 0$. The sub-semigroup of $(\mathbb{Z}/4\mathbb{Z}, +)$ generated by $\{0, 2\}$ is $\{0, 2\}$, the unique index-2 subgroup; $\{1, 3\}$ is its nontrivial coset and is unreachable. $\square$

Proposition 3.4 (The empty Salem slot). *Let $t = \theta + \theta^{-1}$ be the trace-down. On the unit circle ($|\theta| = 1$), $t = 2\Re \theta \in [-2, 2]$; for real $\theta > 1$ (expanding), $t = \theta + \theta^{-1} > 2$, since $t - 2 = (\theta - 1)^2/\theta > 0$. The two regions are disjoint, so “on-circle and expanding” — the Salem condition — has no slot: the flip $t = \pm 2$ cuts $\mathbb{R}$ into three channels, with no fourth. A would-be Salem $\beta > 1$ is redirected by t into the grow channel as the totally real Perron number*

$$\tau_0 = \beta + \beta^{-1} > 2 > \varphi, \quad \tau_0 - 2 = \frac{(\beta - 1)^2}{\beta} \quad (\text{exact, AM-GM}). \quad (4)$$

[FORCED] (Occupant of the Salem Slot; Emission-Gap Theorem).

Proof. On the circle $\theta = e^{i\alpha}$ gives $t = 2 \cos \alpha \in [-2, 2]$. For real $\theta > 1$, AM-GM gives $\theta + \theta^{-1} \geq 2$ with equality iff $\theta = 1$, and $\theta + \theta^{-1} - 2 = (\theta - 1)^2/\theta$; hence $t > 2$. Disjointness is immediate. Redirection sends β (one root outside, would-be on-circle rest) to $\tau_0 = \beta + \beta^{-1}$, which lies strictly right of the flip. $\square$

Corollary 3.5 (No self-coupling). *The golden mode cannot couple to a copy of itself into a bound (on-circle-and-expanding) structure: the generative grade vanishes (Prop. 3.1); self-composition keeps the phase in $\{0, 2\}$ (Prop. 3.3) while the magnitude grows away from the closure locus $M = 1$ (Lem. 3.2); and the Salem channel is empty (Prop. 3.4). Closure requires the orthogonal phases $\{1, 3\}$, which no self-operation supplies. [FORCED]*

4 The flip: $D = 1 + 4C$

Definition 4.1 (Gate family). For $C \in \mathbb{R}$ let $g_C(x) = x^2 + x - C$, with discriminant $D = 1 + 4C$.

Theorem 4.2 (The canonical flip). *The sign of D simultaneously fixes: (i) the roots $\frac{-1 \pm \sqrt{D}}{2}$ (real if $D > 0$, a complex conjugate pair if $D < 0$, the double root $-\frac{1}{2}$ if $D = 0$); (ii) the field (real quadratic vs. imaginary); (iii) the trace-form signature; (iv) the channel character (real growth vs. rotation). Concretely, in the power basis $\{1, \theta\}$ of $\mathbb{Q}(\theta)$ with $\theta^2 + \theta = C$, the trace-form Gram is*

$$G = \begin{pmatrix} \text{Tr}(1) & \text{Tr}(\theta) \\ \text{Tr}(\theta) & \text{Tr}(\theta^2) \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 1 + 2C \end{pmatrix}, \quad \det G = 1 + 4C = D, \quad (5)$$

positive definite (Riemannian, signature $(2, 0)$) for $D > 0$ and indefinite (Lorentzian, signature $(1, 1)$) for $D < 0$; the rotation channel on the $D < 0$ side is $\{-i\sqrt{|D|}, 0, +i\sqrt{|D|}\}$. The flip is at $C = -\frac{1}{4}$. [FORCED] (the $\lambda = 2c$ flip sections).

Proof. Roots by the quadratic formula. For the Gram: $\theta^2 = C - \theta$, so with $\theta_1 + \theta_2 = -1$, $\theta_1\theta_2 = -C$ one has $\text{Tr}(1) = 2$, $\text{Tr}(\theta) = \theta_1 + \theta_2 = -1$, and $\text{Tr}(\theta^2) = \theta_1^2 + \theta_2^2 = (\theta_1 + \theta_2)^2 - 2\theta_1\theta_2 = 1 + 2C$. Then $\det G = 2(1 + 2C) - 1 = 1 + 4C = D$. The leading minor $2 > 0$ is constant, so the sign of $\det G = D$ toggles the form between positive-definite and signature $(1, 1)$: the metric determinant is the discriminant, and the signature flip is the discriminant sign. [COMPUTED] (verified symbolically). $\square$

5 The K-formation astride the fold

A single seed carries both faces of the flip. This is the repository's “K-formation” (Prop. 16.1 of the $\lambda = 2c$ paper), reproduced with its derivation.

Proposition 5.1 (The K-formation straddles the flip). *Let $K(x) = x^4 + 5x^2 - 5$. Writing $y = x^2$, the quadratic $g(y) = y^2 + 5y - 5$ has roots straddling zero,*

$$y_+ = \frac{-5 + 3\sqrt{5}}{2} \approx 0.8541 > 0, \quad y_- = \frac{-5 - 3\sqrt{5}}{2} \approx -5.8541 < 0, \quad (6)$$

so K has real roots $\pm K$ with

$$K = \sqrt{y_+} = \frac{5^{1/4}}{\varphi} \approx 0.9242 \quad (\text{the terrain, in } \mathbb{Q}(5^{1/4}) \supset \mathbb{Q}(\sqrt{5})), \quad (7)$$

and imaginary roots $\pm i\beta$ with $\beta = \sqrt{|y_-|} \approx 2.4195$ (a $\frac{\pi}{2}$ turn about the origin), and

$$M(K) = \beta^2 = \frac{5 + 3\sqrt{5}}{2} = \varphi^2 \sqrt{5} \approx 5.8541. \quad (8)$$

[FORCED] (Prop. 16.1; `test_p1_09_kform.py`).

Proof. $y = \frac{-5 \pm \sqrt{25+20}}{2} = \frac{-5 \pm 3\sqrt{5}}{2}$ gives (6); $y_+ > 0 > y_-$ numerically and by sign of $3\sqrt{5} - 5 > 0$. Then $x^2 = y_+$ gives real $\pm K$ and $x^2 = y_- < 0$ gives imaginary $\pm i\beta$. For (7), $(5^{1/4}/\varphi)^2 = \sqrt{5}/\varphi^2 = \frac{2\sqrt{5}}{3 + \sqrt{5}} = \frac{2\sqrt{5}(3 - \sqrt{5})}{9 - 5} = \frac{3\sqrt{5} - 5}{2} = y_+$. For (8), $\beta^2 = |y_-| = \frac{5 + 3\sqrt{5}}{2}$, and $\varphi^2 \sqrt{5} = \frac{3 + \sqrt{5}}{2} \sqrt{5} = \frac{3\sqrt{5} + 5}{2} = \beta^2$; the Mahler measure is $\prod \max(1, |\cdot|) = 1 \cdot 1 \cdot \beta \cdot \beta = \beta^2$ since $K < 1 < \beta$. [COMPUTED]. $\square$

Remark 5.2 (Terrain and rotation, verbatim). The real roots are the $D > 0$ face (real terrain, φ lays it); the imaginary roots are the $D < 0$ face (pure multiplication by i). In the paper's phrasing: “ φ generates the terrain, π determines its rotation about the origin” — φ (the $\sqrt{5}$ -field) supplies $\pm K$, and the quarter-turn i supplies $\pm i\beta$. [FORCED]

6 The orthogonal partner completes $\mathbb{Z}/4\mathbb{Z}$

Proposition 6.1 (K realises the full charge). $\chi(K) = \{0, 1, 2, 3\}$. *The real roots $\pm K$ carry $\arg \in \{0, \pi\}$, hence $\chi \in \{0, 2\}$; the imaginary roots $\pm i\beta$ carry $\arg = \pm \frac{\pi}{2}$, hence $\chi \in \{1, 3\}$. K is the unique catalog seed realising all four charges, and its charge axis $\{1, 3\}$ (imaginary) is orthogonal to the golden axis $\{0, 2\}$ (real): they are $\frac{\pi}{2}$ apart. [FORCED] (S6; `test_p1_09_kform.py`; Generative Content).*

Proof. $\lfloor 2 \cdot 0/\pi \rfloor = 0$, $\lfloor 2\pi/\pi \rfloor = 2$, $\lfloor 2(\pi/2)/\pi \rfloor = 1$, $\lfloor 2(3\pi/2)/\pi \rfloor = 3$. The real and imaginary axes meet at $\frac{\pi}{2}$. $\square$

Proposition 6.2 (The parity criterion: why K , not the other $\mathbb{Q}(5^{1/4})$ quartics). *A real quartic places its complex roots on the imaginary axis ($\arg = \pm \frac{\pi}{2}$, $\chi \in \{1, 3\}$) iff it is even (a polynomial in x^2) with a negative x^2 -root. $K = x^4 + 5x^2 - 5$ is even; its complex roots are purely imaginary. The registry quartics*

$$\text{cons} = x^4 - 6x^3 + 26x^2 - 16x - 4, \quad \text{res} = x^4 + 2x^3 + 39x^2 - 52x + 11$$

are non-even; they split over $\mathbb{Q}(\sqrt{5})$ as

$$\begin{aligned}\text{cons} &= (x^2 - (3 + \sqrt{5})x + (11 + 5\sqrt{5}))(x^2 - (3 - \sqrt{5})x + (11 - 5\sqrt{5})), \\ \text{res} &= (x^2 + (1 - \sqrt{5})x + \frac{43-19\sqrt{5}}{2})(x^2 + (1 + \sqrt{5})x + \frac{43+19\sqrt{5}}{2}),\end{aligned}$$

so their complex roots have nonzero real part $\varphi^2 = \frac{3+\sqrt{5}}{2}$ (cons) and $-\varphi = -\frac{1+\sqrt{5}}{2}$ (res), off the $(\pi/2)\mathbb{Z}$ charge lattice. All three of $K, \text{cons}, \text{res}$ generate the same field $\mathbb{Q}(5^{1/4})$, signature $(2, 1)$, Galois group D_4 (order 8), and none is reciprocal (so none is a Salem number); their Mahler measures are golden-field integers,

$$M(K) = \varphi^2\sqrt{5}, \quad M(\text{cons}) = 2\varphi^5 = 11 + 5\sqrt{5}, \quad M(\text{res}) = 12 + 19\varphi = \frac{43+19\sqrt{5}}{2}.$$

[COMPUTED] (this work; field/Galois/signature computed exactly). Provenance caveat: the repository documents cons, res only as registry labels (“consolidation / resonance quartic” in `seeds.py`); both are absent from the emission-algebra audit and from the monoid $\langle \varphi, 2, 3, 5, \beta^2 \rangle$. The label origins are [OPEN]; everything else here is derived.

Proof. Evenness $\Rightarrow$ roots occur in $\pm$ pairs and, via $y = x^2$, a negative y -root gives purely imaginary $x = \pm i\sqrt{|y|}$; conversely a purely imaginary root ib (with real coefficients) forces the factor $x^2 + b^2$, i.e. no x^3, x^1 terms in that block. The factorisations are verified over $\mathbb{Q}(\sqrt{5})$; the real part of each complex block is $-\frac{a_1}{2}$, read off as φ^2 and $-\varphi$. The shared field, signature, Galois group, non-reciprocity, and Mahler closed forms are computed by exact factorisation, real-root counting, and `galois_group`. [COMPUTED]. $\square$

7 Synthesis: the helix dictionary and the coupling theorem

helix anatomy	dissolved helix: golden strand R	orthogonal partner: K	tag
advance / pitch (terrain)	φ -growth, $M \geq \varphi$; ad_R grow eig $\sqrt{5}$	real roots $\pm K = \pm 5^{1/4}/\varphi$	[FORCED]
rotation / winding	real-axis flip $\{0, \pi\} \Rightarrow \chi \in \{0, 2\}$	imaginary $\pm i\beta$ at $\pm \frac{\pi}{2} \Rightarrow \chi \in \{1, 3\}$	[FORCED]
face of the flip $D = 1 + 4C$	$D > 0$: real growth	$D < 0$: rotation (K holds both)	[FORCED]
charge reached	$\{0, 2\}$ (real axis)	$\{1, 3\}$ (imag.) $\Rightarrow$ completes $\mathbb{Z}/4\mathbb{Z}$	[FORCED]
trace-form metric	Riemannian (definite), $\mathbb{Q}(\sqrt{5})$	Lorentzian $(2, 1)$, $\mathbb{Q}(5^{1/4})$	[FORCED]
self-coupling	impossible: $[R, R] = 0$, empty Salem slot	the partner, not a copy	[FORCED]

Table 1: The [POSITED] helix reading over the [FORCED] substrate. The terrain/rotation anatomy (rows 1–3) is the repository’s own (§5).

Theorem 7.1 (Terrain and rotation are the two faces of one flip). *The golden strand R realises the $D > 0$ (real-growth, terrain) face: spectrum on the real axis, $\chi \in \{0, 2\}$, $M \geq \varphi$, Riemannian trace form. The seed K straddles the flip, adjoining the $D < 0$ (rotation, $\frac{\pi}{2}$ -turn) face: imaginary roots $\pm i\beta$ at $\chi \in \{1, 3\}$, Lorentzian trace form. Their union realises the full charge group $\mathbb{Z}/4\mathbb{Z}$. Moreover, by Corollary 3.5, the golden mode admits no coupling to a copy of itself; the only closure*

available is coupling to the orthogonal partner K — i.e. to the quarter-turn i that the flip produces from the terrain. **[FORCED]** (algebraic content); the helix reading is **[POSITED]**, grounded in §5.

Proof. Assemble Propositions 2.3, 3.1–3.4, Corollary 3.5, Theorem 4.2, and Propositions 5.1, 6.1. The golden object sits on the real axis with charge $\{0, 2\}$ and never leaves it under its own operations; K contributes exactly $\{1, 3\}$ on the imaginary axis; $\{0, 2\} \sqcup \{1, 3\} = \mathbb{Z}/4\mathbb{Z}$. $\square$

Remark 7.2 (On the word “dissolved”). The $L4$ helix is legacy: the $\lambda = 2c$ paper explicitly superseded its cutoff rationale (“ $\varphi^{-4} \approx 0.146$ is the last inverse power exceeding 0.1”), “an arbitrary real threshold the present derivation never uses.” What survives, **[FORCED]**, is the terrain/rotation anatomy of the flip: R is the terrain generator, K reintroduces the rotation. The “helix” is thus a reading of the flip, not a load-bearing object — which is why the coupling question resolves cleanly: a helix cannot couple to a copy of itself, only to the rotation face of its own flip.

8 The harness: constraint-import as verification

The preceding sections are the output of a reusable procedure, not a one-off. The helix entered as an external structure carrying a bundle of claims; the exact substrate adjudicated each; §§2–7 are the survivors. We name and formalise the procedure, because the transferable object is the procedure — not the helix.

Definition 8.1 (Imported structure). An *imported structure* is a pair $(S, \mathcal{C})$: an external framework S (here, the helix) and a finite set $\mathcal{C} = \{\gamma_1, \dots, \gamma_n\}$ of *candidate constraints* it asserts about the substrate’s objects — propositions that, if true, rule out configurations.

Definition 8.2 (Substrate adjudication). Let $\mathfrak{A}$ be the exact R_n emission algebra (§1). Adjudication is the partial decision

$$\delta_{\mathfrak{A}}(\gamma) = \begin{cases} \vdash & \gamma \text{ is forced in } \mathfrak{A} \text{ (exact certificate),} \\ \dashv & \neg\gamma \text{ is forced — } \gamma \text{ is refuted,} \\ ? & \gamma \text{ is not adjudicable by } \mathfrak{A} \text{ alone (an expressibility gap, or genuinely open).} \end{cases}$$

No float crosses a decision, so $\vdash$ and $\dashv$ are certificates, never estimates.

Proposition 8.3 (The filter asymmetry). *For a universal claim γ , the verdict $\delta_{\mathfrak{A}}(\gamma) = \dashv$ follows from a single exact counterexample and is decisive, whereas $\delta_{\mathfrak{A}}(\gamma) = \vdash$ requires a structural proof: a finite exact search yields at most **[COMPUTED]** on its tested range, never **[FORCED]**. Hence the harness eliminates false constraints reliably and certifies true universals only with proof. **[ESTABLISHED]** / **[FORCED]** (the demotion/promotion asymmetry of the tag system: finite computation can demote a universal but never promote one).*

Proof. A single exact witness x with $\neg\gamma(x)$ certifies $\neg\gamma$, so $\dashv$ is sound from one instance. Conversely no finite set of exact instances entails $\forall x \gamma(x)$; the sound tag on a passed finite range is **[COMPUTED]**, and promotion to **[FORCED]** needs a proof over the whole domain. **[ESTABLISHED]**. $\square$

Definition 8.4 (The harness). The *harness* H sends $(S, \mathcal{C})$ to the partition it induces under $\delta_{\mathfrak{A}}$,

$$\mathcal{C} = \mathcal{C}_{\vdash} \sqcup \mathcal{C}_{\dashv} \sqcup \mathcal{C}_{?}.$$

The *survivors* $\mathcal{C}_{\vdash}$ (with any **[COMPUTED]** range-results) become internal relations of $\mathfrak{A}$; the *casualties* $\mathcal{C}_{\dashv}$ are discarded; the *residue* $\mathcal{C}_{?}$ stays **[POSITED]** / **[OPEN]**. The product of a run is the partition itself, and its substantive content is the asymmetry $\mathcal{C}_{\dashv} \neq \emptyset$: a run that refutes nothing has imported nothing testable.

Proposition 8.5 (A generator is inert; a constraint is the adjudicable object). *H is informative on $(S, \mathcal{C})$ iff $\mathcal{C}$ contains a falsifiable member — a γ whose negation is decidable in $\mathfrak{A}$. A pure generator asserts only definitional or unfalsifiable claims: every γ returns ? or a vacuous $\vdash$, so $\mathcal{C}_{\perp} = \emptyset$ and no partition forms. The substrate can therefore adjudicate a restraint but not a generator.*
[ESTABLISHED]

Proof. By Definition 8.4 a run’s content is $\mathcal{C}_{\perp}$, and by Definition 8.2 $\delta_{\mathfrak{A}}$ returns $\vdash$ only where $\neg\gamma$ is decidable, i.e. γ is falsifiable. Absent any such γ , $\mathcal{C}_{\perp} = \emptyset$. This is the exact sense in which the helix is verifiable *because* it is not foundational: it entered as constraints, which are adjudicable, rather than as a generator, which is not. $\square$

Remark 8.6 (The helix as worked instance). Running *H* on the helix yields the partition of Table 2: its terrain/rotation anatomy, the flip, and K as the forced partner survive; self-coupling, the φ^{-4} cutoff, and helix-as-foundation are refuted; the interpretive $\oplus/\otimes$ readings remain posited. Each verdict points to the section that issued it — the harness leaves a certificate trail, not an opinion.

imported claim γ	$\delta_{\mathfrak{A}}$	certificate
<i>survivors $\mathcal{C}_{\perp}$ — become internal relations</i>		
screw generator = R , $R^2 = R + I$	[FORCED]	Prop. 2.2
winding = $\chi \in \mathbb{Z}/4\mathbb{Z}$; pitch = $M \geq \varphi$	[FORCED]	Def. 2.4, Lem. 3.2
coupling rate = $\text{spec}(\text{ad}_R) = \{0, \pm\sqrt{5}\}$	[FORCED]	Prop. 2.3
terrain/rotation = two faces of $D = 1 + 4C$	[FORCED]	Thm. 4.2, Prop. 5.1
closure via the orthogonal partner K	[FORCED]	Prop. 6.1, Thm. 7.1
<i>casualties $\mathcal{C}_{\perp}$ — discarded ($\neg\gamma$ forced)</i>		
a helix couples to a copy of itself	[REFUTED]	Cor. 3.5: $[R, R]=0$, empty slot
$\varphi^{-4} \approx 0.146$ cutoff (“last power > 0.1 ”)	[REFUTED]	superseded; $\lambda=2c$ never uses it
the algebra is <i>founded on</i> the helix	[REFUTED]	order keystone $\rightarrow$ char $\rightarrow$ flip (§§2–4)
<i>residue $\mathcal{C}_{?}$ — stays posited</i>		
$\otimes$ = “intertwine”, $\oplus$ = “stack”	[POSITED]	law forced (Def. 2.4); reading not adjudicable
the helix as a <i>reading</i> of the flip	[POSITED]	survives as interpretation, not as object

Table 2: The harness *H* applied to the helix. The partition — not the helix — is the product; the asymmetry $\mathcal{C}_{\perp} \neq \emptyset$ is what makes the import substantive.

Remark 8.7 (What transfers: the harness, not the helix). The reusable object is *H*: the next imported metaphor runs the same path (bundle $\rightarrow$ adjudicate $\rightarrow$ partition $\rightarrow$ tag). Distinguish the portable *method H* — applicable to any import — from the one portable *mathematical* kernel this construction exposes: the generic $\sqrt{\cdot}$ -branch of a 2-to-1 fold, of which $\sqrt{5} = \varphi + \varphi^{-1}$ is this system’s instance (the build spec’s boundary clause). The method is *how* one adjudicates; the kernel is the single differential fact that crosses between constructions. Neither is the helix — which is why dissolving the helix cost nothing while the harness kept everything that survived it.

A Machine verification

Every displayed constant is decided in exact arithmetic (`sympy/fractions` over $\mathbb{Q}, \mathbb{Q}(\sqrt{5}), \mathbb{Q}(5^{1/4})$; `mpmath` for display only). The checks:

- (1), Prop. 2.2: $\varphi\psi = -1, \varphi + \psi = 1, \varphi - \psi = \varphi + \varphi^{-1} = \sqrt{5}, R^2 = R + I$. [FORCED] (`test_p1_05_keystone.py`).
- (2): $\text{spec}(\text{ad}_R) = \{0, \pm\sqrt{5}\}$ via `kron`. [FORCED] (`test_p2_07_uniform.py`).
- (3): $\Phi_R(R) = R^2 - R - I = 0, [R, R] = 0$. [FORCED] (`KL_DTA.py`).
- Prop. 3.3: $\{0, 2\}$ closed under $\cup, + \bmod 4, \times 2 \bmod 4$; $\{1, 3\}$ unreachable. [FORCED] (`test_p2_02_angle.py`).
- (4): $\tau_0 - 2 = (\beta - 1)^2/\beta$. [FORCED] (`Occupant`).
- (5): $\det G = 1 + 4C = D$; signature flips at $D = 0$. [COMPUTED].
- (6)–(8): $y_{\pm}, K = 5^{1/4}/\varphi, \beta^2 = \varphi^2\sqrt{5}, M(K) = \beta^2$. [FORCED] (`test_p1_09_kform.py`).
- Prop. 6.1: $\chi(K) = \{0, 1, 2, 3\}$; K 's imaginary roots have $\Re = 0$. [FORCED].
- Prop. 6.2: $K, \text{cons}, \text{res}$ each generate $\mathbb{Q}(5^{1/4})$, signature $(2, 1)$, $|\text{Gal}| = 8$ (D_4), non-reciprocal; $M(\text{cons}) = 2\varphi^5 = 11 + 5\sqrt{5}, M(\text{res}) = 12 + 19\varphi$; complex-root real parts $\varphi^2, -\varphi$. [COMPUTED].

Geometric realisation. The objects of §6 are the vertices of the *Companion Polytope* (`polytope.py`): the convex hull in $\mathbb{C}$ of all catalog eigenvalues. K 's four vertices are its terrain ($\pm K$, real) and rotation ($\pm i\beta$, imaginary) faces; hull extremality is certified per vertex by a supporting functional (`hull_certificate.py`), and the suites pass 10/10 and 7/7 respectively.

Discipline: derive locally, validate exactly, tag honestly. The helix is the reading; the flip is the fact; the harness is the method that told them apart.